

Hardware Implementation of Cell-Free MIMO and Dynamic TDD using the OAI 5G NR Codebase

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Abstract—The cell-edge user in a traditional cellular network experiences degraded quality of service compared to a user at the centre of the cell due to path loss and interference from neighboring base stations (BSs). Cell-free multiple input and multiple output (MIMO) is a promising technology that can improve the cell-edge performance as it offers macro-diversity and consequently improved coverage and throughput. Moreover, the network capacity in a cellular network is often underutilized due to a static time division duplex (TDD) configuration coupled with asymmetric traffic demands in neighboring cells. To handle heterogeneous traffic demands, dynamic TDD is considered to be a beneficial technology, wherein different cells can independently schedule slots in the uplink (UL)/downlink (DL) directions depending on their local traffic demands. However, dynamic TDD brings new challenges because of the introduction of cross-link interferences (CLI) between users and between BSs. Combining cell-free MIMO with dynamic TDD can effectively overcome the CLI, as there exists inbuilt co-operation between the access points (APs) in the cell-free MIMO architecture. This paper presents a hardware implementation of cell-free MIMO and dynamic TDD using the OPENAIRINTERFACE (OAI) 5G new radio (NR) software stack, with the aim to experimentally understand the role of cell-free MIMO and dynamic TDD in providing better connectivity at the cell-edge and improving network throughput.

Index Terms—cell-free MIMO, dynamic TDD, OpenAirInterface, 5G NR, USRP

I. INTRODUCTION

THE cellular network topology has been utilized for many decades to cover a large geographical area with wireless communication services, where each base station (BS) serves an exclusive set of users. The spectral efficiency has been gradually improved by reducing cell sizes, increasing the number of antennas at the BS and users, applying advanced signal processing techniques for interference mitigation, and so on. However, since the received signal power rapidly decays with the propagation distance, the user who happens to be close to the BS experiences a higher signal-to-noise ratio (SNR) than those close to the edge between the two cells. Moreover, users at the cell edge are also affected by interference from neighboring cell BSs. As a result, the signal-to-interference-plus-noise ratio (SINR) becomes substantially lower than the SNR at these locations. The poor performance obtained as a

user moves from the cell-center to the cell-edge degrades the network's service quality as a whole.

Cell-free multiple input and multiple output (MIMO) is a promising technology that can improve the cell-edge performance as it offers macro-diversity and consequently improved coverage and throughput. In cell-free MIMO [1], many access points (APs), each potentially equipped with multiple antennas, geographically distributed over a given area and connected to a central processing unit (CPU) via a fronthaul network, jointly serve all the user equipments (UEs) by coherent transmission and reception. The CPU operates the system in a distributed MIMO fashion, with no cell boundaries. Based on the co-operation among the APs, there exist different levels of abstraction in the radio access network for the cell-free MIMO implementation, such as, centralized processing (CP), distributed/local processing (DP) or partial processing at each AP and at the CPU.

In cell-free MIMO DP, in the downlink (DL), the CPU delivers the information to the APs, which locally process the information and transmit the signal to the UE. In the uplink (UL), the APs receive the signal from the UE, locally process the information to generate soft estimates of the data, which are then forwarded to the CPU for joint detection. In cell-free MIMO CP, in the DL, the CPU processes the information and generates baseband samples which are then forwarded to the APs, where the radio frequency (RF) processing (signal upconversion) is done, and the signal is transmitted to the UE. In the UL, the APs receive the signal from the UE, perform the RF processing (signal downconversion) to generate the baseband samples, which are then forwarded to the CPU for joint detection.

Since the centralized version requires more information to be exchanged between the CPU and the APs, it needs to have larger fronthaul capacity compared to the distributed version. In cell-free MIMO DP, the APs need to be equipped with multiple antennas for mitigating the interference as only local estimates are available for processing. This is unlike cell-free MIMO CP, where the centralized processing can overcome the interference even with single antenna APs.

Apart from poor performance at the cell edge, the network capacity in a cellular network is often underutilized due to a *static* time division duplex (TDD) configuration coupled with asymmetric traffic demands in neighboring cells. In the TDD mode, the UL and DL transmissions use the same carrier frequency and are separated in time only. In static TDD, the same transmission direction (UL or DL) is used across all the cells operating in the same frequency band to reduce inter-cell

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interference. This defers the transmissions until a UL (DL) slot appears, if the primary traffic demand is in the DL (UL) direction. For example, if two neighboring cells have mostly UL and DL demands respectively, then static TDD leads to under-utilization of DL slots in one cell and UL slots in the other.

Dynamic TDD is considered to be a potential solution to meet heterogeneous traffic demands in nearby cells. In dynamic TDD [2], the network can dynamically use resources for either UL or DL transmissions to match the instantaneous traffic demand in each cell, leading to improved user performance compared to the conventional static split of resources between UL and DL. However, dynamic TDD brings new challenges because of the introduction of cross-link interference (CLI). In the UL, the BS receives interference from the BSs in the adjacent DL cells, called DL-to-UL interference or BS-to-BS interference. In the DL, a DL UE receives interference from the UL UEs of the adjacent UL cells, called UL-to-DL interference or UE-to-UE interference.

Combining cell-free MIMO with dynamic TDD can effectively overcome the CLI, as there exists inbuilt co-operation between the APs in the cell-free MIMO architecture. When we integrate cell-free MIMO with dynamic TDD [3], UL reception and DL transmission simultaneously occur over the same band. *This provides in-band full-duplex functionality with a half-duplex hardware complexity, since the BSs collectively act as a full-duplex transceiver.* However, this requires a high level of synchronization among the APs as the distributed nature of the system may result in the problem of intersymbol interference or intercarrier interference, and degrade the system performance.

Many papers in the literature discuss the benefits of either cell-free MIMO or dynamic TDD using simulation. However, only a few of them have attempted to verify the technology's feasibility and efficiency in the context of current 3GPP standards. In [4], cloud based distributed cell-free massive MU-MIMO system is implemented and tested for the 3.5 GHz carrier frequency and bandwidth of 100 MHz with 128×128 antenna system which reveals that the system can support 5G NR compatible transmissions. However, the developed system is restricted for the verification of cell-free architecture with centralized processing and static TDD communications. The design, implementation and practical evaluation of cell-free MIMO and dynamic TDD still remains a challenge.

This paper presents a hardware implementation of cell-free MIMO and dynamic TDD using the OpenAirInterface (OAI) 5G new radio (NR) software stack [5], with the aim to experimentally understand the role of cell-free MIMO and dynamic TDD in providing better connectivity at the cell-edge and improving network throughput. The performance of the system is evaluated at a carrier frequency of 3.5 GHz.

The rest of the paper is organized as follows. Section II details the system architecture and describes the prototype implementation. Section III presents the test results. Section IV concludes the paper and indicates potential directions for future work.

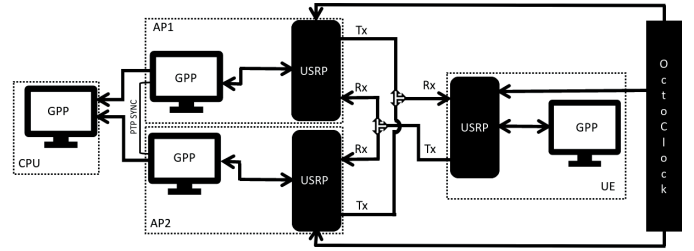


Fig. 1. Block diagram of the cell-free MIMO setup with distributed processing.

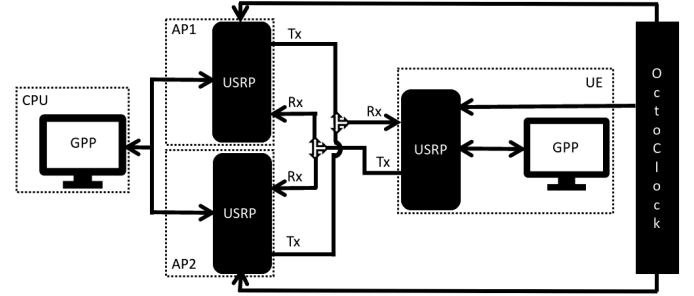


Fig. 2. Block diagram of the cell-free MIMO setup with centralized processing.

II. SYSTEM ARCHITECTURE

A. Cell-free MIMO

For demonstration, we consider two cases of the cell-free MIMO architecture, viz., DP and CP.

1) *Cell-free MIMO DP*: Fig. 1 shows the setup of cell-free MIMO DP. There are two AP general-purpose processors (GPPs), synchronized using IEEE-1588 precision time protocol (PTP). Each AP runs the OAI gNB protocol stack for baseband processing. The AP GPPs communicate to the CPU GPP via the user datagram protocol (UDP). In addition, there is one GPP on which OAI UE protocol stack runs for baseband processing. Finally, a universal software radio peripheral (USRP) X310 is connected to each AP and UE GPP for RF processing and all the USRPs are synchronized using an OctoClock CDA-2990. In each USRP, a transmitter and a receiver is enabled. For UL, the data transmitted by the UE is split using a two-way splitter which is then fed to the receivers of the USRPs of each AP. For data decoding, selection combining (SC) is used, where the cyclic redundancy check (CRC) flag (generated after low-density parity check (LDPC) decoding) stamped with frame number, slot number, and segment number is sent to the CPU by both the APs. If any of the APs' segment has the CRC flag as 1, the segment is declared as CRC successful. In the DL, the APs transmit the data synchronously, and is combined using a two-way combiner. The combined data is then fed to the receiver of the UE.

2) *Cell-free MIMO CP*: Fig. 2 shows the setup of cell-free MIMO CP. There is one CPU GPP that runs the OAI gNB protocol stack for baseband processing, to which two time synchronized AP USRPs are connected for RF processing.

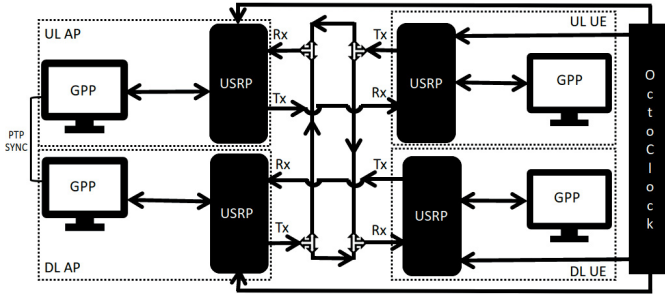


Fig. 3. Block diagram of the dynamic TDD setup.

Another GPP runs the OAI UE protocol stack for baseband processing to which a single USRP is connected for RF processing. In each USRP, a transmitter and a receiver is enabled. In the DL, the synchronous data transmitted from the APs are multiplexed using a two-way combiner, which is then received at the UE. In the UL, the data transmitted from the UE is split using two-way splitter and fed to the receiving APs. The data received at both the APs is sent to the CPU, where it goes through a joint decoding process. For decoding of the UL data, two methods are used, namely, maximum ratio combining (MRC) and SC. In case of MRC, the two streams of data are combined using the individual channel estimates of each of the receiving channels and the combined data is then sent to the LDPC decoder for further processing. In SC, the LDPC decoded data of each AP is combined using the CRC flag, namely, the data decoding is declared successful if the decoded data from at least one AP passes the CRC.

B. Dynamic TDD

For demonstration of dynamic TDD in cellular architecture we consider the setup shown in Fig. 3. There are two time-synchronized AP GPPs, each running the OAI gNB protocol stack for baseband processing. One AP GPP handles DL processing and the other handles UL processing. In addition, there are two UE GPPs which run the OAI UE protocol stack for baseband processing. One UE GPP manages the UE with dominant DL traffic and another manages the UE with dominant UL traffic. To each GPP, a single antenna USRP is connected for RF processing. The UL AP receives the UL traffic as well as the interference from DL AP. The DL UE receives the DL traffic as well as the interference from the UL UE. To study the effect of interference, in one of the slots, both the DL and UL transmissions are configured to happen using the same resource elements.

III. TEST RESULTS

The performance of our cell-free MIMO and dynamic TDD system is measured through the observed packet error rate. The transport block received by the physical layer shared channel, post the CRC attachment, is divided into segments and each segment is again appended with another CRC [6]. The transport block CRC or the segment wise CRC can be used to trigger the hybrid automatic repeat request re-transmissions at the receiver side. In the OAI 5G codebase,

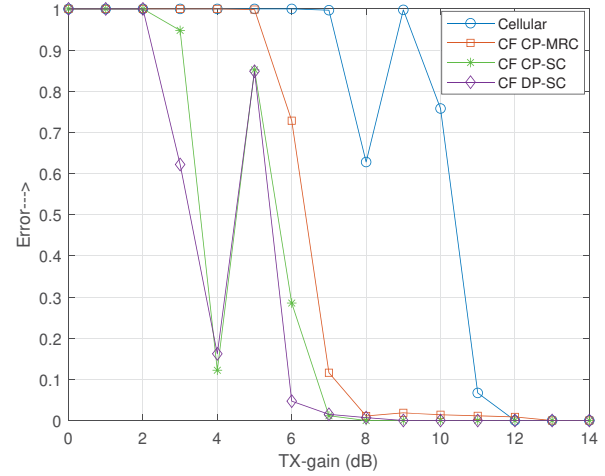


Fig. 4. Packet error rate at CPU versus UE's RF TX gain for the UL cell-free MIMO setup, with MRC and SC in the case of centralized processing, and with SC in the case of distributed processing.

the re-transmissions are disabled and the segment wise CRC is used for checking the accuracy of the received data. If the segment is successfully received, the CRC_OK flag is raised, otherwise the CRC_NOT_OK flag is raised. Using this, the packet error rate is computed as:

$$\text{Error} = 1 - \frac{\text{No. of CRC_OK}}{\text{No. of CRC_OK} + \text{No. of CRC_NOT_OK}}$$

A. Cell-Free MIMO

1) *Uplink*: Fig. 4 shows the packet error rate at the CPU versus UE's RF transmitter (TX) gain for cell-free MIMO UL processing. The transmission/reception in this case is done via SubMiniature version A (SMA) cables with 36 dB attenuation with modulation and coding scheme (MCS) index as 10¹, 50 resource blocks (which corresponds to a bandwidth of 18 MHz), and the receiver (RX) gain for both the APs is set to 5 dB. The packet error rate is averaged over 5 experiments, with 10,000 segments being transmitted in each experiment. For measurements in the cellular case (for comparison), the same setup as for cell-free MIMO is used, except that one of the interfaces of the splitter output is blocked. This is done to ensure that approximately the same signal power loss due to the splitter exists in both the cases. It is observed from the plot that the cell-free MIMO shows an improvement of about 5 dB in UE's RF TX gain for SC based decoding for both DP & CP and 4 dB for MRC based decoding in CP compared to the cellular scenario. This surpasses the theoretically expected improvement of 3 dB, which could be because the USRP only supports a 1 dB granularity in setting the TX gain, and also possibly because of hardware non-idealities, for example, using a splitter with one interface blocked could have led to a greater loss in the signal strength in the cellular case compared to the cell-free case.

¹An MCS index of 10 corresponds to a 16-QAM constellation and a rate 340/1024 LDPC code.

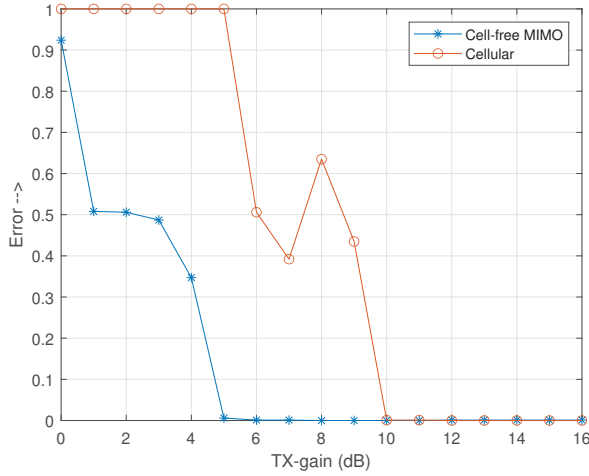


Fig. 5. Minimum packet error rate at the CPU versus the AP's RF TX gain for the DL cell-free MIMO setup.

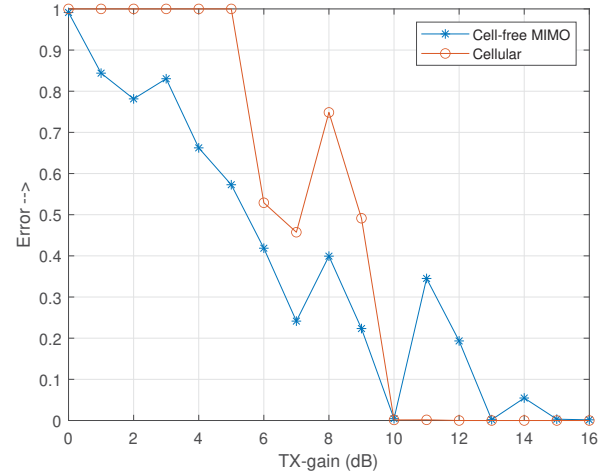


Fig. 6. Average packet error rate at the CPU versus the AP's RF TX gain for the DL cell-free MIMO setup.

2) *Downlink*: Fig. 5 shows the minimum² packet error rate at the CPU versus AP's RF TX gain for cell-free MIMO DL processing. The transmission/reception is done over 50 resource blocks using 36 dB attenuation with the SMA cables and the MCS index of 10. The RX gain for the UE is fixed at 5 dB. In each experiment, 10,000 segments are transmitted and the experiment is repeated 10 times. The same setup is used for the cellular DL measurement, blocking one of the combiner interfaces. In this case, a 5 dB improvement is observed in the cell-free MIMO case as compared to the cellular case. Fig. 6 shows the average packet error rate at CPU versus AP's RF TX gain for cell-free MIMO DL processing. The result shows improvement in cell-free MIMO when the data is transmitted at low TX gains. However, as the TX gain is increased, both the cellular and the cell-free MIMO perform equally well. The reason for cell-free MIMO's performance degradation on increasing the AP's TX gain can be because the data is not precoded, due to which the system cannot leverage the full diversity gain. With the use of Maximum Ratio Transmission (MRT), the average DL performance of cell-free MIMO can be further improved.

B. Dynamic TDD

The lab setup of dynamic TDD is shown in Fig.7.

Fig. 8 shows the packet error rate at the DL UE versus the UL UE's TX gain for the dynamic TDD setup. The transmission/reception in this case is done via SMA cables with 36 dB attenuators. We use 10 dB attenuators between the splitter and combiner in the interfering path to avoid coupling effects. The data is transmitted over 50 resource blocks with an MCS index of 10. The RX gain of the DL UE is fixed at 10 dB, and the DL AP's TX gain is set to 8 dB. In each experiment, 5,000 segments are transmitted and the packet error rate is averaged over 5 experiments. It is observed from the plot that when the UL UE is transmitting, the error is high compared

to when the UL UE is off. Also, as the UL UE's TX gain is increased, the interference is increased, and, hence, the error rate is also increased. This illustrates the effect of the UE-to-UE interference.

Fig. 9 shows the packet error rate at the UL AP versus the DL AP's TX gain for the dynamic TDD setup. The transmission/reception in this case also is done via SMA cables with 36 dB attenuators. As before, 10 dB attenuators are used between the splitter and combiner in the interfering path to avoid coupling. The data is transmitted over 50 resource blocks with an MCS index of 5.³ The RX gain of the UL AP is fixed at 15 dB, and the UL UE's TX gain is set to 8 dB. The packet error rate is averaged over 5 experiments, with 5,000 segments being transmitted in each experiment. It is observed from the plot that when DL AP is transmitting, the error is high compared to when DL AP is off. This depicts the AP-to-AP interference. The increase in interference with increase in DL AP's TX gain is not monotonic, which can possibly be because of hardware non-idealities.

IV. CONCLUSION

This paper described the design and implementation of the cell-free MIMO architecture and dynamic TDD, in a practical experimental setup with 5G NR compliant signaling. Experimental results demonstrating the advantages of both technologies were presented, in terms of the packet error rate at the receiver. These results elucidated the potential performance improvement that can be obtained in real-world systems.

The work can be extended further to establish and analyze the performance of an integrated setup comprising of cell-free MIMO and dynamic TDD. Also, the system can be extended to an end-to-end 5G NR system and support various research works and industry collaborations.

³An MCS index of 5 corresponds to a QPSK constellation and a rate 379/1024 LDPC code.

²Minimum value obtained among the experiments carried.

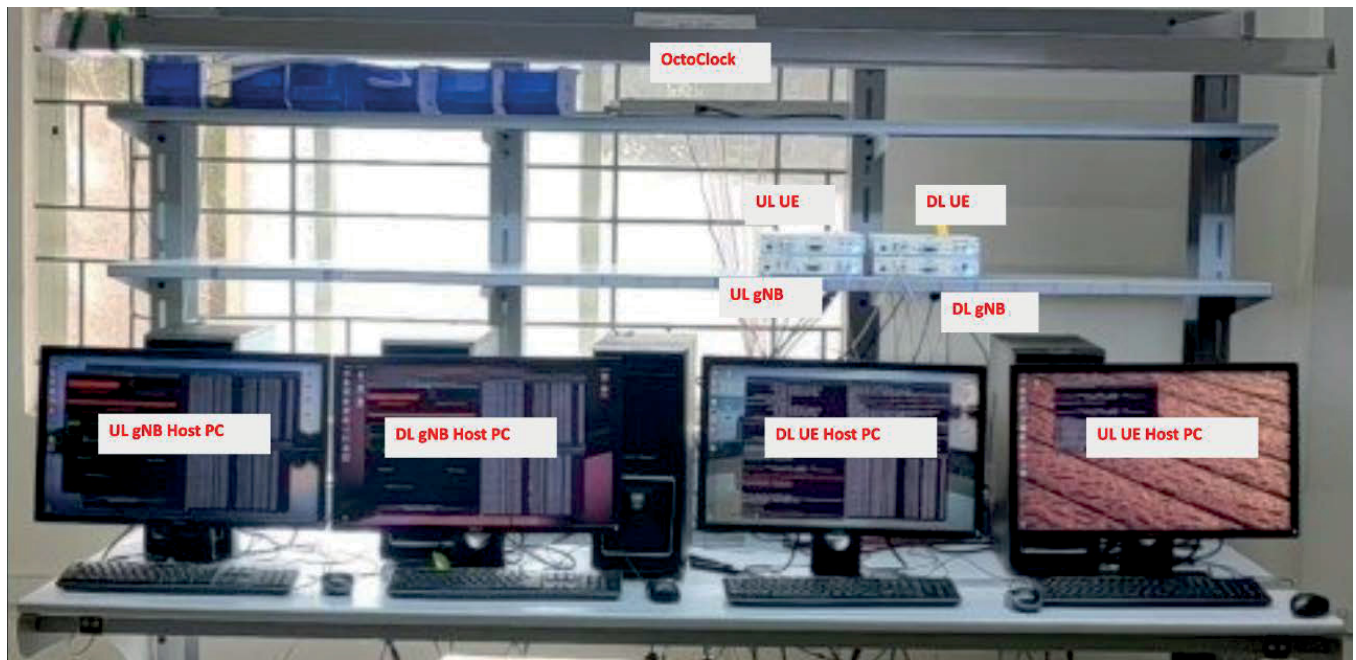


Fig. 7. Dynamic TDD lab setup.

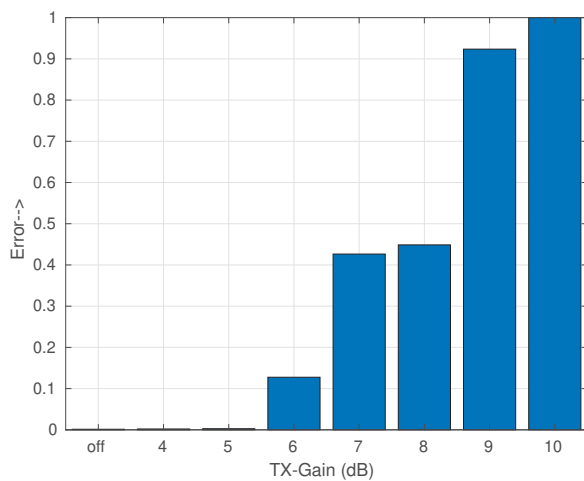


Fig. 8. Average packet error rate at the DL UE versus the UL UE's RF TX gain for the dynamic TDD setup.

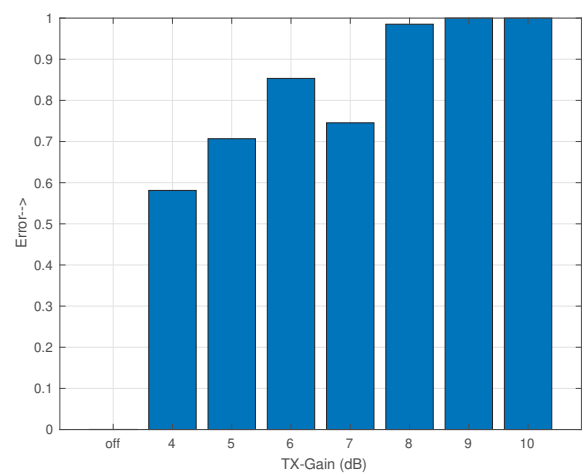


Fig. 9. Average packet error rate at the UL AP versus the DL AP's RF TX gain for the dynamic TDD setup.

REFERENCES

- [1] E. Björnson and L. Sanguinetti, "Making cell-free massive MIMO competitive with MMSE processing and centralized implementation," *IEEE Transactions on Wireless Communications*, vol. 19, no. 1, pp. 77–90, 2020.
- [2] H. Kim, J. Kim, and D. Hong, "Dynamic TDD systems for 5G and beyond: A survey of cross-link interference mitigation," *IEEE Communications Surveys Tutorials*, vol. 22, no. 4, pp. 2315–2348, 2020.
- [3] J. Li, Q. Lv, P. Zhu, D. Wang, J. Wang, and X. You, "Network-assisted full-duplex distributed massive MIMO systems with beamforming training based CSI estimation," *IEEE Transactions on Wireless Communications*, vol. 20, no. 4, pp. 2190–2204, 2021.
- [4] D. Wang, C. Zhang, Y. Du, J. Zhao, M. Jiang, and X. You, "Implementation of a cloud-based cell-free distributed massive MIMO system," *IEEE Communications Magazine*, vol. 58, no. 8, pp. 61–67, 2020.
- [5] OpenAirInterface Software Alliance. [Online]. Available: <https://gitlab.eurecom.fr/oai/openairinterface5g/-/tree/develop>
- [6] "5G NR: Multiplexing and channel coding," 3rd Generation Partnership Project (3GPP), Technical Specification (TS) 38.212, version 15.10.0.